

PROJECT ABSTRACT REPORT

Azure Chaos Studio Resilience Experiment

TEAM NUMBER: 189

PROJECT NUMBER: 087

PROBLEM STATEMENT:

Modern cloud applications are expected to remain available and recover quickly even when infrastructure failures occur. However, designing a highly available architecture alone does not guarantee that the system can withstand real-world failures. This project addresses the need to **validate application resilience by using Azure Chaos Studio to inject a controlled zone failure**. The experiment evaluates how the application responds to the failure, whether traffic is successfully handled by redundant resources, and how quickly the system recovers. It also identifies challenges related to experiment scoping, permissions, agent configuration, and monitoring while ensuring that the failure test is performed safely without affecting unrelated resources.

ABSTRACT:

This project is being done as a part of the AZ-104 Microsoft Azure Administrator certification course. The goal of the project is to demonstrate how to solve practical problems with cloud services using the Microsoft Azure platform.

Our team aims to study the given problem, identify the required Microsoft Azure services, and devise and build a solution with those services. The project's primary goal is to gain a deeper understanding of how to design, implement, and manage a solution on the cloud.

Instead of relying solely on theoretical concepts, this project will take a hands-on approach. It will explore various services on Azure to understand how they are combined to provide a working solution on the cloud. As we progress through the project, we will analyze issues we encounter, research possible solutions and try to apply them step by step.

OBJECTIVES:

- Understanding the requirements for the given project.
- Understanding various Microsoft Azure services.
- Identifying the cloud resources to be used for the implementation.
- Designing a suitable cloud architecture.
- Gaining hands-on experience in managing the cloud resources.

USE CASES:

- How applications can be hosted in the cloud.
- Managing Virtual Machines in Microsoft Azure.

- Understanding cloud storage solutions in Microsoft Azure.
- Managing users and accessing to the resources.
- Monitoring resources and application health.
- Security features and backup services.

REQUIREMENTS:

Software Requirements

- Microsoft Azure Portal
- Web Browser
- Microsoft Azure Account Access
- Internet Connection

Hardware Requirements

- Personal Computer/ Laptop
- Minimum 8 GB RAM
- Internet Connectivity

METHODOLOGY:

- Step 1: Study the project description and understand the expected outcome.
- Step 2: Analyze the problem and identify the Azure services that may be required.
- Step 3: Prepare a basic design for the proposed solution.
- Step 4: Create the necessary resources in the Azure environment.
- Step 5: Test the implementation and observe the results.
- Step 6: Document the findings and make changes if required.

EXPECTED OUTCOMES:

- Better understanding of cloud computing concepts.
- Practical exposure to Microsoft Azure services.
- Improved knowledge of cloud resource management.
- Better understanding of designing cloud-based solutions.

CONCLUSION:

This project is currently in its initial stages, and our team is focusing on understanding the project scope and initial design planning. Through a structured and sequential approach, we aspire to acquire hands-on experience with Microsoft Azure and gain a comprehensive understanding of its real-world applicability.

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